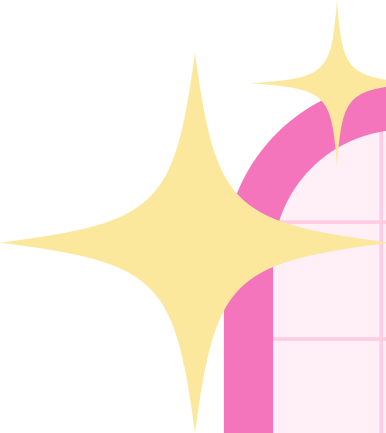
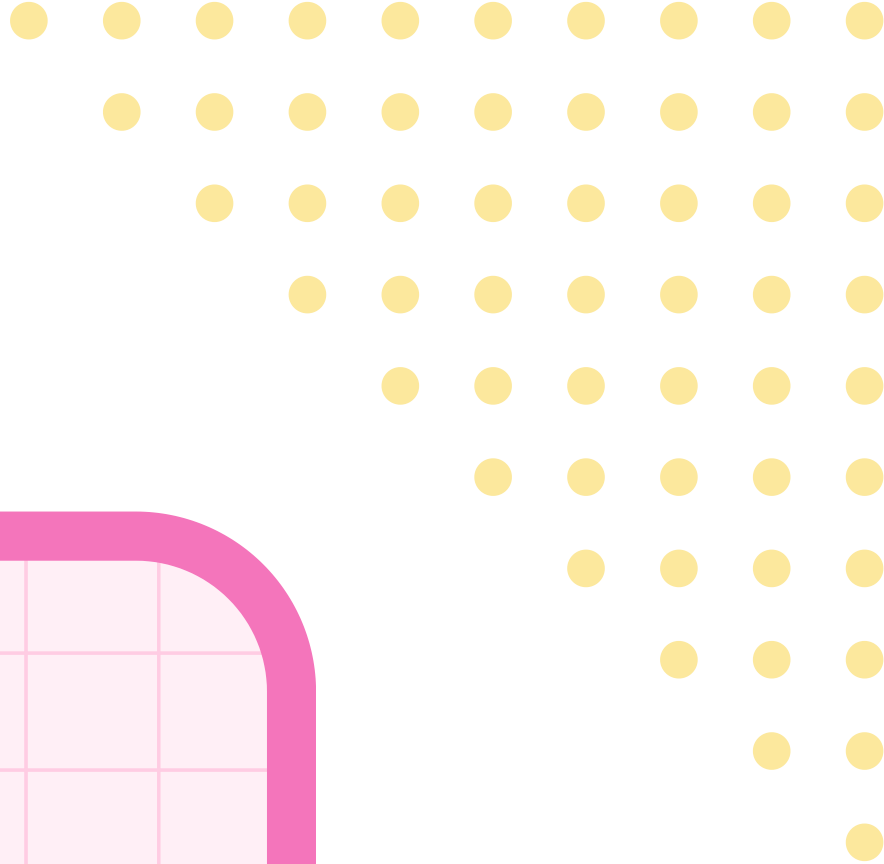
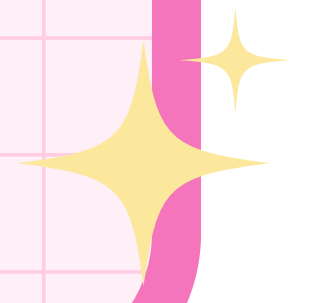




Research-Oriented University System

Object-Oriented Programming in Java • Console-Based Application

TOTALLY SPIES!



Mission Accomplished



Authentication

Login / logout, role-based menu routing for all 5 user types



Course Registration

Credit (max 21) & fail-count (max 3) constraints enforced



Mark System

$\text{att1} \times 0.3 + \text{att2} \times 0.3 + \text{final} \times 0.4 \rightarrow$
letter grade A-F



Transcript

GPA on 4.0 scale with formatted academic record



Research Module

Papers sorted by date / citations / length via Strategy pattern



Supervisor h-index

LowHIndexException if professor
 $\text{h-index} < 3$



ResearchProject

NonResearcherJoinException on
invalid participant



Singleton DB

Java Serialization — data persists
between runs



UserFactory

Admin creates any role through a
single Factory call



Manager

Approve registrations, assign
teachers, generate reports



Admin

Add / remove / ban users, reset
passwords, view logs



Researcher Ranking

Top-cited per school, sorted by
citations / date / length

Mission Critical

★ Teacher-Rating Display

Rating data stored but not surfaced in Student menu — view method exists but not wired to menu router.

🔄 Circular Serialization Edge Case

Student→Transcript→Mark→Student cycle causes occasional NullPointerException on deserialization.

📋 Admin Log Viewer — No Filtering

Reads raw log file in full — no filter by date, user, or action. Unusable on large files.

🏆 Cross-University Ranking — Year Filter Bug

Off-by-one in year filter excludes boundary-year researchers from cross-university sort.

🖨️ Input Validation on Scanner Reads

Scanner.nextInt() not wrapped in try-catch — non-integer input crashes menu loop.

Gadget Upgrade Protocol

★ Teacher-Rating Display

Wire `TeacherRating.getAverage()` into Student menu — one switch-case entry + loop over `course.getInstructors()`. ~20 lines.

↻ Circular Serialization Edge Case

Mark Transcript as transient in Student, add custom `readObject()` to rebuild from mark list. Or switch to Gson with `@Expose`.

📋 Admin Log Viewer — No Filtering

Load lines into `List<String>`, filter with `Stream.filter()` by keyword/user/date. Add simple sub-menu: by user / by date / show all.

🏆 Cross-University Ranking — Year Filter Bug

Change `'year < targetYear'` → `'year <= targetYear'` in stream filter. Add unit test with boundary-year researcher to pin the fix.

⌨️ Input Validation on Scanner Reads

Wrap `Scanner.nextInt()` in try-catch (`InputMismatchException`), call `scanner.nextLine()` to flush, re-prompt in `while(true)` loop.

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Custom
Exceptions

Sabit Amina

User Hierarchy & Infrastructure
User · Employee · Admin ·
Manager
Database (Singleton) ·
AuthService
UserFactory · Enums ·
Exceptions

Turkpen Adiya

Academic Module
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SeniorLecturer
Student · Course · Lesson ·
Mark
Transcript · CourseRegistration
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Nurbek Anzara

Research & Integration
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ResearchPaper
ResearchProject · Strategy
pattern
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